

Preliminary Research Note

Problem framing, evidence map, assumptions, and early design direction

PROJECT

Integrated Robotic Actuator

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Research Question

Which actuator architecture gives the best path to a compact, safe, and controllable robotic joint prototype?

Abstract

This document captures early research for the Integrated Robotic Actuator project. It is intended to separate observations, assumptions, risks, and design direction before committing to a detailed technical design.

Keywords: integrated actuator, robotics, torque density, control architecture, preliminary research

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1 RESEARCH SNAPSHOT

Research Snapshot

Problem	Build a compact integrated robotic actuator that combines motor, reduction, sensing, drive electronics, and control firmware.
Key Finding	The highest leverage early decision is the actuator architecture, because it constrains torque density, controllability, thermal behaviour, and manufacturability.
Main Uncertainty	The true operating envelope is still uncertain: load profile, duty cycle, heat rejection, encoder resolution, and gearbox backlash all need validation.
Next Step	Prototype the smallest credible bench rig and measure torque, speed, temperature rise, backlash, and closed-loop response.

2 PROBLEM FRAMING

Research Question – System Direction

What architecture can demonstrate a credible portfolio-grade actuator while still being feasible with undergraduate-level tools, budget, and testing equipment?

The preliminary research phase should not over-specify the product. Its job is to reduce uncertainty and identify the design direction that is most likely to produce a demonstrable robotic actuator prototype.

Hypothesis – Architecture

A modular integrated actuator with a BLDC motor, compact reduction stage, magnetic encoder, current sensing, and embedded field-oriented control is likely to provide the best balance between technical ambition and achievable implementation.

3 EVIDENCE MAP

Table 1: Evidence Matrix

Claim	Evidence / Observation	Source	Confidence
Integrated packaging is valuable	Combining actuation, sensing, and drive control reduces wiring complexity and makes the unit easier to reuse across robots.	Project observation	Medium
Backlash matters	Low backlash is important for repeatable joint control and credible closed-loop demonstrations.	Bench-test requirement	High
Thermal limits may dominate	Small actuators can reach thermal limits before electrical or mechanical limits during sustained load.	Engineering assumption	Medium

4 ASSUMPTIONS TO VALIDATE

Table 2: Assumptions to Validate

Assumption	Why it matters	Validation method	Owner
Motor torque is sufficient	Sets whether the reduction ratio and motor frame size are credible.	Measure no-load speed and loaded torque on a bench rig.	Controls
Encoder resolution is adequate	Limits position accuracy and velocity estimation quality.	Compare commanded and measured step response.	Firmware
Thermal rise is manageable	Determines continuous operating limits.	Run fixed-duty thermal test with logging.	Electrical

5 OPEN GAPS

Open Gap – Unknowns

The research note should explicitly track what is not known yet: final motor choice, gearbox type, torque target, power supply envelope, controller current limit, and safety cut-off design.

Risk – Prototype Risk

The main early risk is selecting parts before the test envelope is clear. Keep the first rig instrumented and conservative, especially around power electronics and rotating hardware.

6 RESEARCH BACKLOG

Table 3: Research Backlog

ID	Question / Task	Output	Priority
R-001	Compare candidate actuator architectures: direct drive, belt reduction, planetary gearbox, cycloidal concept.	Architecture trade study	High
R-002	Build a first-order torque-speed and thermal model.	Notebook with plots	High
R-003	Define safety plan for power bring-up and moving-mechanism tests.	Bring-up checklist	High
R-004	Identify demo scenarios for the final portfolio video.	Demo storyboard	Medium

7 PRELIMINARY DIRECTION

Direction / Decision – Next Iteration

Proceed toward a bench-testable integrated actuator module. The next document should convert this research into measurable requirements, architecture diagrams, and a verification plan.